UNIT TEST PLAN

Project Name: **FETCH: Inventory Management System with Predictive Analytics for Risha Pet Supplies**

Date:

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Run/Tested by:

Legend:

- **Passed (Result is same as expected)**
- **Failed (Actual result is NOT same as expected result)**
- **In Progress (Test is still in progress or not yet done)**
- **Open (Not yet tested)**

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
ALL ROLES — ADMINISTRATOR, CASHIER AND SUPPLIER							
Login Module							
	1	Completeness Check					
			Check if a valid sign in works: enter the username and the password, then click Sign in	The user is signed in and taken to the home page of the account			
			Check if Username is empty: leave Username empty, then click Sign in	The request is not sent and the tooltip "Please fill out this field." is displayed on Username			
			Check if Password is empty: leave Password empty, then click Sign in	A tooltip "Please fill out this field." is displayed on Password			
			Check if the reset username or email is empty: type the username, click "Forgot your password?", clear the Username or email box, then click Ask for a reset	A tooltip "Please fill out this field." is displayed			
			Check if New password is empty: open a reset link, leave New password empty, then click Save new password	A tooltip "Please fill out this field." is displayed			
			Check if a username is needed for a reset: leave Username empty, then click "Forgot your password?"	A message "Enter your username first." is displayed and the reset box does not open			
	2	Format Check					
			Check if a password with every kind of character is accepted: open a reset link, enter a new password with an uppercase letter, a lowercase letter, a number and a special character, then click Save new password	A message "Password has been reset. You can now sign in. Redirecting to sign in..." is displayed			
			Check if the password has no uppercase letter: enter a new password without an uppercase letter	A message "Password must contain at least one uppercase letter (A–Z)" is displayed			
			Check if the password has no lowercase letter:	A message "Password must contain at least one			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			enter a new password without a lowercase letter	lowercase letter (a–z)" is displayed			
			Check if the password has no number: enter a new password without a number	"Password must contain at least one number (0–9)" is displayed and users.password is not updated			
			Check if the password has no special character: enter a new password without a special character	A message "Password must contain at least one special character (e.g. !@#$%)" is displayed			
	3	Range Check					
			Check if a password of 6 or more characters is saved: enter a new password of 6 characters or more that follows the rules, then click Save new password	The password is saved and the user is taken to the Sign in page			
			Check if the password is shorter than 6 characters: enter a new password with fewer than 6 characters	A message "Password must be at least 6 characters long" is displayed			
			Check if a 6th wrong sign-in is blocked: enter a wrong password 5 times within 15 minutes, then try a 6th time	POST /api/auth/login returns HTTP 429 with "Too many login attempts. Try again in (seconds)s." and no session token is issued			
	4	Consistency Check					
			Check if the two passwords do not match: open a reset link, enter a new password and a different password in Confirm password, then click Save new password	"Passwords do not match." is displayed and users.password is not changed			
	5	Database Check					
			Check if a saved account can sign in: enter a username and password that are saved in the system, then click Sign in	POST /api/auth/login returns HTTP 200 with a signed token holding the user id, role and token_version; the token cookie is set and the role's home page opens			
			Check if the password is wrong: enter a username with a wrong password, then click Sign in	POST /api/auth/login returns HTTP 401 "Invalid username or password" (the same answer as an unknown username), no token cookie is set and the failure is counted toward the lockout			
			Check if the email is not yet confirmed: sign in with an account whose email is not yet confirmed	POST /api/auth/login returns HTTP 403 because users.email_verified is 0: "Please confirm your email address first. Check (email) for the confirmation link." is displayed and no token is issued			
			Check if the account is deactivated: sign in with an account that was deactivated	The account has users.is_active = FALSE, so it is answered like an unknown username: HTTP 401 "Invalid username or password" and no token is issued			
			Check if the reset link has expired or was used:	No users row has that reset_token_hash with			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			open a reset link that has expired or was already used	reset_token_expires still ahead, so HTTP 400 "This reset link is invalid or has expired. Please request a new one." is displayed and users.password is not changed			
	6	System Check					
			Check if the Administrator opens the Dashboard: sign in as Administrator	HTTP 200, the token cookie is set and dashboard.html opens with the shop's figures from GET /api/dashboard/stats			
			Check if the Cashier opens the Point of Sale: sign in as Cashier	The Point of Sale opens, not the Dashboard, with "Cashier: (name)" taken from the signed-in user's full_name			
			Check if the Supplier opens the Supplier Portal: sign in as Supplier	supplier.html (Supplier Portal) opens and lists only the purchase orders and products whose supplier_id is that supplier			
			Check if "Keep me signed in" keeps the session: tick "Keep me signed in", sign in, close the browser and open FETCH again	The token is still valid, so the user is signed in without entering the password again			
ADMINISTRATOR							
Navigation Module							
	1	Consistency Check					
			Check if each menu opens its own records: sign in as Administrator and click Dashboard, Inventory, Sales, Suppliers, Reports and Users in the sidebar one after another, then click the bell and View all notifications	Each request returns HTTP 200 and each page lists its own records: products, sales, suppliers, user accounts and notifications			
			Check if the bell lists the latest notices: click the Notifications bell	A list opens beside the bell with the latest notices from the Notifications page, and the number on the bell shows how many are unread			
			Check if a cashier can open Inventory: sign in as Cashier, then type the address of the Inventory page (inventory.html)	The server redirects the cashier to pos.html, and Inventory API calls from a cashier account are refused with HTTP 403			
			Check if a supplier can open the Dashboard: sign in as Supplier, then type the address of the Dashboard (dashboard.html)	The server redirects the supplier to supplier.html (Supplier Portal), and shop API calls from a supplier account are refused with HTTP 403			
			Check if the logo hides and shows the menu: click the Risha logo at the top of the sidebar (it has a small arrow badge), then click the logo again	The sidebar narrows to icons and the logo shows "Show the menu" with a right-pointing arrow; clicking it shows the full sidebar again			
	2	System Check					

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			Check if Jump to finds a product: press Ctrl + J (Cmd + J on a Mac), type the name of a product, then press Enter	The product is listed among the matches (5 at most) and Enter opens that product			
Dashboard Module							
	1	Consistency Check					
			Check if the product counts match Inventory: compare Total Products, Restock Needed and Expiring Soon with the matching figures in Inventory	The numbers are the same on both pages (both count the same product records)			
			Check if Total Revenue matches the Sales page: compare Total Revenue with Total Sales Amount on the Sales page	The amounts are the same (both count the same sales)			
			Check if "Within 30 days" lists only those products: click the "Within 30 days" card in the Expiring products panel	Only products whose expiration date is within the next 30 days are listed			
			Check if Needs restocking lists only low products: look at the Needs restocking list	Only products at or below their reorder level are listed			
			Check if a well-stocked product is left out: look at the Needs restocking list for a product whose stock is above its reorder level	That product is not listed			
	2	System Check					
			Check if a voided sale is left out of revenue: void a sale, then look at Recent Sales and Total Revenue	The sale reads "Voided" in Recent Sales and Total Revenue goes down by its final_amount			
Inventory Module							
	1	Completeness Check					
			Check if a complete product is added: click Add, fill in Product name and Category, choose the Product type, fill in Unit price and Supplier, then click Save product	A message "Product added" is displayed with the SKU it was given (for example DRY-121), and the product is in the list			
			Check if Product type is not chosen: click Add, leave Product type unselected, then click Save product	A message "Choose the product type. The SKU is made from it." is displayed under Product type, and the cursor goes to the first field that needs fixing			
			Check if Product name is empty: click Add, leave Product name empty, then click Save product	"Enter the product name." is displayed under Product name and no row is inserted in products			
			Check if Category is not chosen: click Add, leave Category unselected, then click Save product	A message "Choose a category from the list." is displayed under Category			
			Check if Unit price is empty: click Add, leave Unit price empty, then click Save product	A message "Enter the unit price (the selling price)." is displayed under Unit price			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			Check if Supplier is not chosen: click Add, leave Supplier unselected, then click Save product	A message "Choose a supplier. A product needs one so it can be reordered." is displayed under Supplier			
			Check if the bulk quantity is empty: tick a product, click Adjust stock, clear the quantity, then click Continue	A message "Type how many, like 10." is displayed and the window stays open			
			Check if the barcode is empty: click Barcodes, leave a barcode empty, then press Enter	A message "Enter a barcode first" is displayed			
	2	Format Check					
			Check if a correct price and stock are saved: click Add, choose a Product type, enter a price such as 120.50 and a whole number in Stock on hand, then click Save product	The product is saved with those values			
			Check if letters can be typed in Unit price: click Add and type letters in Unit price	The letters are not accepted: the field holds numbers only			
			Check if letters can be typed in Stock on hand: click Add and type letters in Stock on hand	The letters are not typed: Stock on hand takes numbers only			
			Check if the SKU can be typed in: click Add and try to type in the SKU field	Nothing can be typed: the SKU field is read-only and is filled in from the Product type			
			Check if the barcode has a space: click Add, enter a barcode with a space, for example "48 00", then leave the field	A message "A barcode is 8, 12 or 13 digits: numbers only, with no letters, spaces or dashes." is displayed under Barcode			
			Check if the price has more than 2 decimals: click Add, enter 120.555 in Unit price, then leave the field	A message "Use at most 2 decimal places, for example 120.50." is displayed under Unit price			
			Check if stock sold by the piece has a decimal: click Add, keep Sold by on Piece and enter 2.5 in Stock on hand, then leave the field	A message "Stock sold by the piece has to be a whole number." is displayed			
			Check if the picture is not a PNG, JPG, WEBP or GIF: click Add, open More details, click Upload and choose a file that is not a picture	A message "Choose a PNG, JPG, WEBP or GIF picture." is displayed			
			Check if the barcode is not 8, 12 or 13 digits: click Add, type 1111 in Barcode, then leave the field	A message "A barcode is 8, 12 or 13 digits. This one has 4." is displayed under Barcode			
			Check if the CSV has no sku or name column: click Import and choose a CSV file without the sku or name column	A message "Missing required columns: sku, name" is displayed			
			Check if the CSV has the header row only: click Import and choose a CSV file with the header row only	A message "CSV must have a header row and at least one data row" is displayed			
	3	Range Check					

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			Check if removing stock within what is on hand works: tick a product with 20 in stock, click Adjust stock, choose Remove stock, enter 5, click Continue, then Yes, Adjust	A message "Stock adjusted" is displayed with "Updated 1 of 1 selected product." and the stock is now 15			
			Check if expired stock is replaced when an order is received: in Suppliers, open the Orders view and click Receive into stock on an order of 6 for an expired product with 20 in stock, enter the new batch's expiration date, then confirm	The 20 expired are written off, the stock becomes 6 and the product takes the new expiration date			
			Check if the product name is too short: click Add, enter A as the Product name, then leave the field	A message "The product name is too short: use at least 2 characters." is displayed			
			Check if Stock on hand is below 0: click Add and enter -5 in Stock on hand, then leave the field	A message "The stock on hand cannot be less than 0." is displayed			
			Check if Reorder level is below 0: click Add and enter -1 in Reorder level, then leave the field	A message "The reorder level cannot be less than 0." is displayed			
			Check if Cost price is above Unit price: click Add, enter 120.50 as the Unit price and 150 as the Cost price, then leave the field	A message "The cost price (₱150.00) is higher than the unit price (₱120.50): the shop would lose money on each sale. Check both prices." is displayed			
			Check if Max stock level is not above Reorder level: click Add, enter 10 as the Reorder level, open More details and enter 5 as the Max stock level	A message "The max stock level has to be higher than the reorder level (10)." is displayed			
			Check if Unit price is too large: click Add and enter 100000000 in Unit price, then leave the field	A message "The unit price is too large (₱99,999,999.99 at most)." is displayed			
			Check if the expiration date is less than 7 days away: click Add and type today's date in Expiration date	A message "The expiration date must be at least 7 days from today: (date) or later." is displayed under Expiration date as soon as the date is typed			
			Check if the bulk quantity is 0: tick a product, click Adjust stock, enter 0, then click Continue	A message "Type a number above 0." is displayed and the window stays open			
			Check if more stock is removed than is on hand: tick a product with 20 in stock, click Adjust stock, choose Remove stock, enter 999, click Continue, then Yes, Adjust	The result lists the product under "Not changed" with "Only 20 in stock: you cannot remove 999." and the stock stays 20			
			Check if the bulk quantity is not a whole number: tick two products, click Adjust stock, type 10+10, then click Continue	A message ""10+10" is not a whole number. Type one number only, like 10 (no + or - signs, no letters). Use Add stock or Remove stock above to choose which." is displayed			
			Check if the picture is larger than 10 MB: click	A message "This picture is (n) MB. Choose one of 10			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			Add, open More details, click Upload and choose a picture larger than 10 MB	MB or less." is displayed and the picture is not added			
	4	Consistency Check					
			Check if Low stock lists only low products: click the Low stock button	Only products with some stock left at or below their reorder level are listed (products at 0 are under Out of stock), and the count on the button equals the number listed			
			Check if Out of Stock lists only products at 0: click the Out of Stock chip	Only products with 0 in stock are listed			
			Check if Expired lists only expired products: click the Expired chip	Only products whose expiration date has passed are listed			
			Check if the category and species filters match: choose a category in All Categories and a species in All Species	Only products of that category and species are listed			
			Check if the search lists only matching products: type a name, SKU or barcode in the search box	Only the products that match are listed			
			Check if the Restock needed button lists low and out of stock: click the Restock needed button	Only products at or below their reorder level are listed, low and out of stock together, and the count on the button equals the number listed			
			Check if the bulk adjustment says what will happen: tick a product, click Adjust stock, choose Add stock and type 5	The line under the quantity reads "Each product gets 5 more."			
			Check if Expired leaves out unexpired products: click the Expired chip	Products that have not expired are not listed			
			Check if a search with no match says so: type a name no product has in the search box	"No products match your filters" is displayed			
	5	Database Check					
			Check if a new product gets the next SKU: click Add, choose a Product type and enter a name and a barcode no other product has, then click Save product	A products row is inserted with sku = the prefix and the highest number in use + 1 (for example DRY-121; deleted products still count, so a number is never given twice), the name and barcode typed, and is_active = TRUE			
			Check if a product edit is saved: open a product, click Edit, change the Unit price, then click Save product	PUT /api/products/(id) updates products.unit_price, an audit_logs row (update, products) keeps the old and new values, and "Product updated" is displayed			
			Check if a product is deleted: click the Delete (trash) button on a product, then click Yes, Delete	products.is_active is set to FALSE (the row and its sales history stay), so the product leaves Inventory, the till and the reorder list; "Product deleted" is displayed			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			Check if the SKU can be changed: open a saved product, click Edit and look at the SKU	The SKU field is read-only, and PUT /api/products/(id) with a different sku is refused with HTTP 400 "The SKU cannot be changed once the product is saved. It stays (sku)."			
			Check if the product name is already used: click Add and enter a Product name another product already has, then click Save product	POST /api/products finds an active product with the same name and answers HTTP 409; the question "This name is already in use" names that product and its SKU, and no row is inserted unless the user chooses to keep both			
			Check if the barcode is already used: click Barcodes, tick Show all, type a barcode that another product already uses, then press Enter	HTTP 409 with "The barcode (code) is already used by (product) (SKU (sku)). Each product needs its own barcode." and products.barcode is not changed			
			Check if Reorder is clicked when nothing is low: click Reorder when no product is at or below its reorder level	GET /api/products/low-stock returns no product with stock_quantity at or below reorder_level, so "No products need reordering" is displayed and no purchase_orders row is inserted			
			Check if Save product is clicked twice: click Save product twice quickly	The Save button reads "Saving…" and ignores a second press while POST /api/products is on its way, so only one products row is inserted			
	6	System Check					
			Check if a deleted product leaves the till: delete a product, then look for it at the Point of Sale	The product is no longer in the till or the reorder list			
			Check if reordering creates purchase orders: click Reorder, tick the products, click Reorder Selected, then click Yes, Create Orders	A message "Purchase orders sent" is displayed with how many orders were made and the suppliers they were emailed to; any notes are listed in the same message			
			Check if expired stock is written off: in Suppliers, open the Orders view, click Receive into stock on an order for a product that has expired, enter the new batch's expiration date, then confirm	The expired stock is written off, the received quantity is added, and the new expiration date is set so the product can be sold again			
Sales History Module							
	1	Completeness Check					
			Check if a date range filters the sales: choose a From and a To date, then click Filter	Only the sales between those dates are listed			
			Check if no date is chosen: click Filter without choosing any date	A message "Select a date range" is displayed			
			Check if the void reason is empty: click Void on a sale, leave the reason empty, then click Void Sale	"A reason is required to void a sale" is displayed and sales.payment_status stays 'completed'			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			Check if the cash counted is empty: click End of day, leave the cash counted empty, then click Save count	A message "Cash counted is not a valid amount" is displayed			
	2	Range Check					
			Check if a count of 0 or more is accepted: click End of day, enter 0 or more as the cash counted	The line under it shows "Balanced" with a tick, "Over by ₱(amount)" or "Short by ₱(amount)"			
			Check if From is later than To: choose a From date later than the To date, then click Filter	"The start date is after the end date" is displayed and no sales are listed for the reversed range			
			Check if the cash counted is below 0: click End of day and type -1000 as the cash counted	The line under it reads "Enter what was counted (0 or more)"			
			Check if the From date is before the first sale: type a From date before the first sale recorded in FETCH, then click Filter	A message "The first sale in FETCH was on (date). Choose that day or later." is displayed			
			Check if the sales date is in the future: type a To date after today, then click Filter	A message "Choose today or an earlier date: there are no sales in the future." is displayed			
			Check if the end-of-day date is in the future: click End of day and type a Business date after today	A message "You cannot count a day that has not happened yet. Choose today or an earlier date." is displayed and the date goes back to today			
	3	Consistency Check					
			Check if only the sales in the range are listed: choose a From and a To date, then click Filter	Only the sales between those dates are listed			
			Check if a sale number finds that sale: type a sale number in the search box	Only that sale is listed			
			Check if Sale Details match the list: click View on a sale	The Sale Details window shows the same items and total as that sale's row			
			Check if Total Sales Amount matches the Dashboard: compare Total Sales Amount with Total Revenue on the Dashboard	The amounts are the same			
			Check if a voided sale is left out of the total: look at a voided sale in the list	Its row reads Voided and its final_amount is not part of Total Sales Amount, which sums only sales that are not voided			
			Check if a voided sale can be voided again: look at a sale that is already voided	It has no Void button, so it cannot be voided twice			
	4	Database Check					
			Check if the sales are exported: click Export CSV while sales are listed	The sales listed are written to a CSV file that downloads, and "Sales exported successfully" is displayed			
			Check if a sale is deleted: click the Delete (trash) button in a sale's row, then click Yes, Delete	DELETE /api/sales/(id) runs as one transaction: for a sale that was not voided each item's stock_quantity			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
				goes back up with an 'in' stock_movements row (reference sale_delete), then its sale_items and sales rows are deleted; an audit_logs row (delete, sales) keeps a copy			
			Check if the end-of-day count is saved: click End of day, enter the cash counted, then click Save count	A cash_reconciliations row is saved with expected_cash, counted_cash and discrepancy (counted − expected), shown as Balanced, Over by or Short by			
			Check if there are no sales to export: click Export CSV when no sale is listed	"No sales to export" is displayed and no file is made			
	5	System Check					
			Check if a sale is voided: click Void on a sale, enter the reason, then click Void Sale	sales.payment_status becomes 'voided' with the reason added to notes, each item's stock_quantity goes back up with an 'in' stock_movements row (reference 'sale_void'), an audit_logs row is written, and "Sale voided" is displayed			
Suppliers Module							
	1	Completeness Check					
			Check if a complete supplier is added: click Add supplier, fill in Supplier name, Contact person, Email, Phone and Payment terms, then click Save supplier	A message "Supplier added" is displayed and the supplier is in the list			
			Check if Supplier name is empty: click Add supplier, leave Supplier name empty, then click Save supplier	The tooltip "Please fill out this field." is displayed on Supplier name and no row is inserted in suppliers			
			Check if Contact person is empty: click Add supplier, leave Contact person empty, then click Save supplier	A tooltip "Please fill out this field." is displayed on Contact person			
			Check if Email is empty: click Add supplier, leave Email empty, then click Save supplier	A tooltip "Please fill out this field." is displayed on Email			
			Check if Phone is empty: click Add supplier, leave Phone empty, then click Save supplier	A tooltip "Please fill out this field." is displayed on Phone			
			Check if the order note is empty: in the Orders view, click Notes on an order and click Send with the note empty	A tooltip "Please fill out this field." is displayed			
			Check if the date for an expired product is empty: in the Orders view, click Receive into stock on an order with an expired product and leave the date empty	A message "(product) (expired (date)) has already expired, so enter the new batch's expiration date" is shown; clicking Receive into stock with no date shows "Enter the new batch's expiration date: the			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
				current one has already passed." and the order is not received			
	2	Format Check					
			Check if a correct email and phone are saved: click Add supplier, enter an email with @ and a phone number in digits, then click Save supplier	The supplier is saved			
			Check if the email has no @: enter an email without @, then click Save supplier	The tooltip "Please include an '@' in the email address." is displayed and the supplier is not saved			
			Check if the phone number has letters: enter a phone number with letters, for example "abc-defg", then click Save supplier	A message "Enter the phone number in digits (at least 7)." is displayed			
	3	Range Check					
			Check if a phone number of 7 digits or more is saved: enter a phone number of 7 digits or more, then click Save supplier	The supplier is saved			
			Check if the phone number has fewer than 7 digits: enter a phone number of fewer than 7 digits, for example 12345, then click Save supplier	A message "Enter the phone number in digits (at least 7)." is displayed			
			Check if the expiration date is less than 7 days away: in the Orders view, click Receive into stock and enter an expiration date less than 7 days from today	A message "The expiration date must be at least 7 days from today: (date) or later." is displayed and the order is not received			
	4	Consistency Check					
			Check if the Rating matches Performance: look at the Rating column, then click Performance on the same supplier	The stars and the result (Meets the standard / Below the standard / Not rated yet) are the same in both places			
			Check if the order history is the supplier's own: click Performance on a supplier	The Purchase order history lists only that supplier's orders			
			Check if each rating condition shows its figure: click Performance on a supplier that has received orders	Each condition is ticked or crossed with its figure: at least 1 order received; at least 75% on time; at least 90% on time; at least 90% in full; average delivery of 7 days or less. 4 stars or more meets the standard			
			Check if each Orders section lists its own orders: open the Orders view	Each section lists only its own orders: Orders on the way, Offered by suppliers, Price changes to decide, and Delivered, not yet paid			
			Check if the search lists only that supplier: type a supplier's name in the search box	Only that supplier is listed			
			Check if a late supplier is below the standard: click Performance on a supplier whose orders all	The on-time conditions are crossed and "Below the standard: it needs 4 stars or more." is displayed			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			arrived after the promised date				
			Check if a supplier with no orders is not rated: click Performance on a supplier with no order received in the last 12 months	A message "Not rated yet: no order from this supplier was received in the last 12 months." is displayed			
	5	Database Check					
			Check if a new supplier is saved: click Add supplier and enter a name that is not on the list, then click Save supplier	POST /api/suppliers inserts a suppliers row with the name typed and is_active = TRUE, and the supplier is listed			
			Check if a supplier edit is saved: click Edit on a supplier, change the phone, then click Save supplier	PUT /api/suppliers/(id) updates suppliers.phone and "Supplier updated" is displayed			
			Check if a supplier is deleted: click Delete on a supplier, then click Yes, Delete	suppliers.is_active is set to FALSE (its orders and products keep their history), the supplier leaves the list and "Supplier deleted" is displayed			
			Check if an order is marked paid: in "Delivered, not yet paid", click Mark paid, then click Mark paid	PUT /api/purchase-orders/(id)/payment sets purchase_orders.paid_at to now; "Marked as paid." is displayed and the order leaves "Delivered, not yet paid"			
			Check if the supplier name is already used: click Add supplier and enter a name that is already on the list, then click Save supplier	HTTP 409 with "A supplier with this name is already on the list." and no second row is inserted			
	6	System Check					
			Check if receiving an order adds the stock: open the Orders view, click Receive into stock on an order, then confirm	purchase_orders.status becomes 'received' with received_at set, each product's stock_quantity goes up by the quantity received, and a stock_movements row 'purchase' (reference 'purchase_order') is added for each item			
Sales Report Module							
	1	Completeness Check					
			Check if a custom period is applied: click Custom, choose a From and a To date, then click Apply	The figures, charts and tables show only that period			
			Check if the custom dates are empty: click Custom, clear both dates, then click Apply	"Choose both a start and an end date" is displayed and no report is produced			
	2	Range Check					
			Check if From on or before To is accepted: click Custom, choose a From date on or before the To date, then click Apply	The report shows that period			
			Check if From is later than To: click Custom,	"The start date is after the end date" is displayed			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			choose a From date later than the To date, then click Apply	and no report is produced			
	3	Consistency Check					
			Check if each period shows only its own sales: click Last 7 days, Last 30 days, This month and All time	Each time, all figures, charts and tables show only the sales of that period			
			Check if Day, Week and Month group the same sales: click Day, Week and Month	The revenue chart and table group the same sales by day, week or month			
			Check if the category lines add up to Revenue: choose All time and look at the lines under Sales by category	"Items sold, all categories" + "Sales recorded without items" − "Discounts given" equals "Revenue", the sum of sales.final_amount for sales that are not voided			
	4	Database Check					
			Check if the sales report is exported: click Export CSV for a period with sales	The report for the period is written to a CSV file that downloads, and "Sales report exported" is displayed			
			Check if the period has no sales to export: click Export CSV for a period with no sales	"There is nothing to export for this period" is displayed and no file is made			
	5	System Check					
			Check if a voided sale is left out of revenue: void a sale, then look at Revenue for that day	Revenue for that day goes down by the voided sale's final_amount			
Stock Planning Module							
	1	Consistency Check					
			Check if Stock planning shows all its parts: click the Stock planning tab	The 30-day forecast, What to reorder, Too much stock and How good is the forecast are displayed			
			Check if each sales-speed filter matches: in All products, click Selling fast, Selling steadily, Selling slowly and Not selling	Each filter lists only the products with that sales speed, with the count on each			
			Check if the detail is that product's own: click a product in All products	The detail shows that product's own sales history and 30-day forecast			
			Check if What to reorder lists only products to order: look at the What to reorder table	Only products that need ordering are listed, each with the reason, stock on hand, on order, order-by date and suggested quantity			
			Check if a product with little sales history is flagged: click a product with fewer than 14 days of sales on record	The detail says "This product has only (n) days of sales on record, so this is a cautious first estimate"			
	2	Database Check					
			Check if the reorder list is exported: click Export list when products need reordering	reorder_list_(YYYY-MM-DD).csv is downloaded with the columns Product, SKU, Supplier, Status, Why, In			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
				stock, On order … Suggested order, and "Reorder list exported" is displayed			
			Check if there is nothing to reorder: click Export list when no product needs reordering	"Nothing needs reordering right now" is displayed and no file is made			
			Check if no product on the list has a supplier: open What to reorder when no product on the list has a supplier	No product on the list has the supplier_id of a current supplier, so the Create purchase orders button is not shown and no purchase_orders row can be made			
Void Requests and Activity Log Module							
	1	Consistency Check					
			Check if only the void requests are listed: click the Void requests tab	Waiting for a decision lists only the sales cashiers asked to void, with the reason and amount			
			Check if a requested sale is kept: click Keep sale on a request, then confirm	"Request dismissed. The sale stands." is displayed and the request leaves Waiting for a decision			
			Check if the activity log filters match: click the Activity log tab, then choose an action and an area	Only audit_logs rows with that action and that table_name are listed			
			Check if there are no void requests: open Waiting for a decision when no cashier has asked for a void	No request is listed			
	2	System Check					
			Check if a requested sale is voided: click Void sale on a request, then confirm	sales.payment_status becomes 'voided', the items go back into stock_quantity with 'in' stock_movements rows, "Sale voided and its items returned to stock" is displayed and the request leaves Waiting for a decision			
Notifications Module							
	1	Consistency Check					
			Check if Low Stock lists only low-stock notices: choose Low Stock in All Types	Only low-stock notices are listed			
			Check if Unread lists only unread notices: choose Unread in All Status	Only unread notices are listed			
			Check if a notice is marked read: click Mark as read (the check button) beside an unread notice	It is marked read and the unread count goes down by one			
			Check if Mark all read marks every notice: click	"All notifications marked as read" is displayed, every			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			Mark all read	notice shows as read and the unread count on the bell is 0			
			Check if Unread is empty after Mark all read: choose Unread in All Status after clicking Mark all read	No notice is listed			
User Management Module							
	1	Completeness Check					
			Check if a complete user is created: click Add user, enter the email, click Send code, enter the 6-digit code and every other field, then click Create user	A message "User created" is displayed and the user is in the list			
			Check if Full name is empty: click Add user, leave Full name empty, then click Create user	The tooltip "Please fill out this field." is displayed on Full name and no row is inserted in users			
			Check if Username is empty: leave Username empty, then click Create user	A tooltip "Please fill out this field." is displayed on Username			
			Check if Email is empty: leave Email empty, then click Send code	A message "Email required" is displayed with "Type the new user's email address first, then send the code."			
			Check if Password is empty: leave Password empty, then click Create user	A tooltip "Please fill out this field." is displayed on Password			
			Check if the supplier is not chosen: choose the role Supplier without choosing a supplier, then click Create user	A message "Supplier required" is displayed			
	2	Format Check					
			Check if a correct email and password are accepted: enter an email with @ and a password with an uppercase letter, a number and a special character	The account can be created			
			Check if the email has no @: enter an email without @	A tooltip "Please include an '@' in the email address." is displayed			
			Check if the password has no uppercase letter: enter a password without an uppercase letter, then click Create user	A message "Password too weak" is displayed with "Add at least one uppercase letter (A–Z)."			
			Check if the password has no number: enter a password without a number, then click Create user	"Password too weak" is displayed with "Add at least one number (0–9)." and no row is inserted in users			
			Check if the password has no special character:	A message "Password too weak" is displayed with			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			enter a password without a special character, then click Create user	"Add at least one special character (e.g. !@#$%)."			
	3	Range Check					
			Check if a password of 6 or more characters is accepted: enter a password of 6 characters or more	The password is accepted			
			Check if the password is shorter than 6 characters: enter a password with fewer than 6 characters, then click Create user	The tooltip "Please lengthen this text to 6 characters or more (you are currently using (number) characters)." is displayed on Password and the account is not created			
			Check if the code is wrong too many times: enter a wrong code 5 times, then try once more	The 5th wrong code shows "That code is not correct. Send a new code." and the next try shows "Too many incorrect attempts. Send a new code."			
	4	Consistency Check					
			Check if the search lists only matching users: type a name in the search box	Only the matching users are listed			
			Check if Show deactivated adds those accounts: tick Show deactivated	Deactivated accounts are added to the list, each with a Restore button			
			Check if deactivated accounts are hidden: untick Show deactivated	Deactivated accounts are not listed			
			Check if you can deactivate your own account: look at your own row in the Users list	There is no Deactivate button on your own row, so an Administrator cannot lock themselves out			
			Check if the email is changed after the code is sent: send a code, then change the email address before creating the user	The line "Step 1: press "Send code" to email a code to that address." is displayed and Create user cannot be clicked			
	5	Database Check					
			Check if a new user is saved: enter a username and an email no other account uses, with the code from the email	A users row is inserted with the username, email and role, a bcrypt hash in password, is_active = TRUE and email_verified = 1 (the code proved the email)			
			Check if an account is restored: tick Show deactivated, then click Restore on a deactivated user	users.is_active is set back to TRUE, "Account restored" is displayed and the user can sign in again			
			Check if the username is already taken: enter a username that is already taken	HTTP 400 with "Username already taken" and no row is inserted in users			
			Check if the email is already used: enter an email that another account already uses	"That email address is already used by (username)." is displayed and no users row is inserted			
			Check if the code is wrong: enter a wrong 6-digit code, then click Create user	The code does not match the code_hash in email_verification_codes: attempts goes up by 1			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
				and "That code is not correct. (n) attempt(s) left." is displayed			
			Check if the code has expired: enter a code that has expired	email_verification_codes.expires_at has passed, so "That code has expired. Send a new one." is displayed and no users row is inserted			
	6	System Check					
			Check if an account is deactivated: click Deactivate on another user, then click Yes, deactivate	"Account deactivated" is displayed, that user's open sessions stop working at their next request, and a new sign-in is refused with "Invalid username or password"			
Settings Module							
	1	Completeness Check					
			Check if a complete profile is saved: in Profile, fill in Full name and Email, then click Save changes	A message "Profile updated" is displayed			
			Check if Full name is empty: in Profile, clear Full name, then click Save changes	"Please enter your full name" is displayed and users.full_name is not changed			
			Check if Current password is empty: in Password, leave Current password empty, then click Update password	A message "Enter current password" is displayed			
			Check if New password is empty: in Password, leave New password empty, then click Update password	A message "Enter new password" is displayed			
			Check if Confirm new password is empty: in Password, leave Confirm new password empty, then click Update password	A message "Confirm new password" is displayed			
	2	Format Check					
			Check if a correct email is saved: in Profile, enter an email with @, then click Save changes	The profile is saved			
			Check if the email has no @: in Profile, enter an email without @, then click Save changes	"Please enter a valid email address" is displayed and users.email is not changed			
			Check if the new password is too weak: in Password, enter a new password without a number or special character	A message "Password too weak" is displayed with the missing rule			
			Check if the photo is not a PNG, JPG, WEBP or GIF: in Profile, upload a file that is not an image as the photo	A message "Choose a PNG, JPG, WEBP or GIF picture." is displayed			
	3	Range Check					
			Check if a discount limit from 0 to 100 is saved: in Preferences, enter 10 in "Largest discount a	A message "Cashiers can now discount up to 10%" is displayed			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			cashier can give (%)", then click Save limit				
			Check if the discount limit is above 100: enter 150 in "Largest discount a cashier can give (%)", then click Save limit	"Enter a limit between 0 and 100" is displayed and the saved limit does not change			
			Check if the photo is larger than 10 MB: in Profile, click Upload photo and choose a picture larger than 10 MB	A message "This picture is (n) MB. Choose one of 10 MB or less." is displayed and the photo does not change			
	4	Consistency Check					
			Check if Session shows the signed-in user: click the Session tab	The full name, role and account status of the signed-in user are displayed (the Cashier role reads "Staff")			
			Check if Profile shows the signed-in user: click the Profile tab	The full name and email of the signed-in user are displayed			
			Check if the two passwords do not match: in Password, enter a different password in Confirm new password	"Passwords do not match" is displayed and the password is not changed			
	5	Database Check					
			Check if a new email is saved: in Profile, enter an email no other account uses, then click Save changes	PUT /api/auth/profile updates users.email after checking no other account has it, and the profile is saved			
			Check if the appearance is saved: in Appearance, choose a mode, accent colour, font and text size, then click Save changes	The mode, accent colour, font and text size are kept in this browser's storage (localStorage) and applied at once; "Appearance saved" is displayed			
			Check if the current password is wrong: in Password, enter a wrong current password, then click Update password	HTTP 400: "Password not changed" is displayed with "Current password is incorrect" and users.password is not changed			
			Check if the email is already used: in Profile, enter an email another account uses	HTTP 400 "That email is already used by another account" and users.email is not changed			
	6	System Check					
			Check if the password is changed: in Password, enter the current password and the same new password twice, then click Update password	users.password is replaced by a new bcrypt hash and token_version goes up (other sessions are signed out); "Password changed" is displayed and the new password works at the next sign-in			
CASHIER							
Point of Sale Module							
	1	Completeness Check					
			Check if the starting cash is saved: on the first sign-in of the day, enter 500 as the Starting cash,	A message "Starting cash saved: ₱500.00" is displayed			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			then click Save starting cash				
			Check if Starting cash is empty: leave Starting cash empty, then click Save starting cash	A message "Enter the cash in the drawer, or 0 if it is empty" is displayed			
			Check if the cart is empty: click Charge with an empty cart	"Cart is empty" is displayed and no sale is recorded			
			Check if Cash received is empty: add an item, leave Cash received empty, then click Charge	A message "Enter the cash amount received" is displayed			
			Check if the cash amount is empty: click Cash in / out, leave Amount empty, then click Record cash taken out	A message "Enter the amount" is displayed			
			Check if the reason for cash taken out is empty: choose Taken out, enter an amount and leave "Who took it, and what for" empty	A message "Say who took the cash and what for" is displayed			
			Check if the cash counted is empty: click End of day, leave the cash counted empty, then click Save count and close drawer	A message "Enter the cash counted in the drawer" is displayed			
			Check if the void reason is empty: in My sales today, click Ask for void and leave the reason empty	A message "Say briefly what went wrong" is displayed			
	2	Format Check					
			Check if numbers in cash and discount are worked out: type numbers in Cash received and Discount	The change and the total are worked out			
			Check if letters can be typed in Cash received: type letters in Cash received	The letters are not accepted: the field holds numbers only			
			Check if letters can be typed in Discount: type letters in Discount	The letters are not typed: the field takes numbers only			
	3	Range Check					
			Check if enough cash is received: enter Cash received equal to or more than the total, then click Charge	The Confirm cash sale window shows the change			
			Check if the quantity is more than the stock: change an item's quantity in the cart to more than its "(n) left"	"Only (n) in stock" is displayed and the quantity is set to n; the server also refuses more than stock_quantity with HTTP 400			
			Check if the cash received is less than the total: enter less Cash received than the total, then click Charge	A message "Insufficient cash — customer gave less than the total" is displayed			
			Check if the discount is above the limit: enter a discount above the limit, for example 50	A message "The largest discount allowed is (limit)%" is displayed and the discount is lowered to the limit			
			Check if an expired product is scanned: scan a	"(product) has expired and cannot be sold" is			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			product whose expiration date has passed	displayed and the product is not added to the cart; POST /api/sales also refuses it with HTTP 400			
			Check if a sold-out product is scanned: scan a product with no stock	A message "(product) is sold out" is displayed			
			Check if more cash is taken out than the drawer holds: take out more cash than the drawer holds	A message "That is more cash than the drawer should hold." is displayed			
	4	Consistency Check					
			Check if the product card matches Inventory: look at a product card	Its price and "(n) left" are the same as the product's Unit price and stock in Inventory			
			Check if a category shows only its products: click a category button	Only the products of that category are shown			
			Check if the search shows only matching products: type a product name in the search box	Only the matching products are shown			
			Check if the subtotal and total add up: add two items and a discount	Subtotal = the sum of quantity × unit price of the lines; Total = Subtotal less the discount %. The server works out the same figures and saves them in sales.total_amount and sales.final_amount			
			Check if the receipt matches the sale: look at the receipt after Complete sale	The items, quantities, discount, total, cash received and change are the same as in the Confirm cash sale window			
			Check if My sales today shows only this cashier's sales: click My sales today	Only this cashier's sales of today are listed			
			Check if the cash in and out totals add up: click Cash in / out	The totals line reads "Put in ₱(amount) · Taken out ₱(amount)" for this drawer today			
			Check if the expected cash adds up: click End of day, type the counted cash, then save the count	The expected cash is not shown before saving (a blind count); after saving it equals starting cash + cash sales + cash put in − cash taken out			
			Check if "Can't be sold" lists only those products: click "Can't be sold"	Only expired and sold-out products are listed, each marked "Expired (date)" or "None left"			
			Check if an unsellable product is among the cards: look for an expired or sold-out product among the product cards	It is not among the cards; it is listed under "Can't be sold"			
			Check if a deleted product leaves the till: delete a product in Inventory while the till is open, then wait 10 seconds	The product's card disappears from the till			
			Check if another cashier's sales are listed: click My sales today after another cashier has made sales	Only your own sales are listed, not the other cashier's			
			Check if a voided sale is left out of the expected cash: void one of today's sales (administrator),	The voided sale is not counted in the expected cash			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			then click End of day at the till				
	5	Database Check					
			Check if a barcode on record adds the product: scan the barcode of a product on record	The barcode matches products.barcode of a product that can be sold, so it is added to the cart with its unit price and "(n) left"			
			Check if a sale is completed: add items, enter Cash received, click Charge, then click Complete sale	A sales row (payment_status completed) and its sale_items rows are inserted, each products.stock_quantity goes down with an 'out' stock_movements row, "Sale completed! Change: ₱(amount)" is displayed and the receipt is shown			
			Check if cash taken out is recorded: click Cash in / out, choose Taken out, enter the amount and reason, click Record cash taken out, then Record it	A till_cash_moves row (kind out, amount, reason) is inserted for today and this cashier; "Recorded" is displayed and the expected cash goes down by that amount			
			Check if the end-of-day count is saved: click End of day, enter the cash counted, then click Save count and close drawer	A cash_reconciliations row is saved for today and this cashier with expected_cash, counted_cash and discrepancy, shown as "Balanced", "Over by ₱(amount)" or "Short by ₱(amount)"			
			Check if the barcode is not on record: scan or type a barcode that is not on record	"No product found for "(barcode)"" is displayed and nothing is added to the cart			
			Check if a void can be asked for twice: look at a sale that already has a void request in My sales today	sales.void_requested_at is already set for that sale, so the Ask for void button is not shown			
			Check if cash can be recorded after the count: after the count is saved, click Cash in / out	A cash_reconciliations row for today exists, so the window says "Today's cash count is already recorded, so nothing more can be added for today." and POST /api/sales/till/cash is refused with HTTP 400			
			Check if a sale can be made after the count: after the count is saved, click a product card	A cash_reconciliations row for today and this cashier exists, so POST /api/sales is refused with HTTP 400 (drawer closed) and "Your drawer is closed for today" is displayed			
	6	System Check					
			Check if the card updates by itself: change a product's stock in Inventory while the till is open, then wait 10 seconds	The product's card shows the new "(n) left" without reloading the page			
			Check if a sale lowers the stock: complete a sale of 2 of a product, then look at its card and at Inventory	A sales row and a sale_items row (quantity 2) are inserted, products.stock_quantity goes down by 2 with a stock_movements row 'out' (−2, reference			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
				'sale'), and the card's "(n) left" and Inventory show the same new stock			
			Check if the sale matches in Sales: sign in as Administrator and open the same sale in Sales	It shows the same items, total and cashier as the receipt			
			Check if Complete sale is clicked twice: click Complete sale twice quickly	Only one sales row is inserted (the second request gets back the sale already saved) and stock_quantity goes down once			
SUPPLIER							
Supplier Portal Module							
	1	Completeness Check					
			Check if an order is confirmed: click Confirm order on a waiting order, choose the delivery date, then click Confirm order	A message "Order confirmed." is displayed			
			Check if the delivery date is empty: click Confirm order and leave the delivery date empty	"Choose the date you will deliver" is displayed and purchase_orders.status stays 'pending'			
			Check if the new delivery date is empty: click Change delivery date and leave the new date empty, then click Save new date	A message "Choose the new delivery date" is displayed			
			Check if the reason for the new date is empty: enter the new date but leave the reason empty, then click Save new date	A message "Say briefly why the date changed" is displayed			
			Check if the offer has no quantity: click Offer an order without entering any quantity, then click Send offer	A message "Enter a quantity for at least one product" is displayed			
			Check if the offer's delivery date is empty: enter a quantity but leave the delivery date empty, then click Send offer	A message "Choose the date you could deliver" is displayed			
			Check if the note is empty: click Notes and click Send with the note empty	A tooltip "Please fill out this field." is displayed			
	2	Format Check					
			Check if a correct receipt number is accepted: click Mark as shipped, enter a receipt number with letters and numbers, then click Mark as shipped	A message "Order marked as shipped." is displayed			
			Check if the receipt number has * or @: enter a receipt number with * or @	"The receipt number can use letters, numbers, spaces and . / # _ - only." is displayed; the order stays 'confirmed' and delivery_ref is not saved			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			Check if the phone number has letters: click Edit contact details and enter a phone number with letters, then click Save details	A message "Enter a phone number using digits, spaces and + ( ) - only." is displayed			
	3	Range Check					
			Check if a price above zero is sent: click Change price, enter a New price greater than zero, then click Send to the shop	A message "Sent to the shop for approval. The current price stays until they agree." is displayed			
			Check if the price is 0: click Change price and enter 0	A message "Enter a price greater than zero" is displayed			
			Check if the delivery date is in the past: click Confirm order and type a delivery date in the past	"The delivery date cannot be in the past." is displayed and purchase_orders.status stays 'pending'			
			Check if the delivery date is over a year away: click Confirm order and choose a date more than a year away	A message "That delivery date is more than a year away." is displayed			
			Check if 5 offers are already waiting: click Offer an order when 5 offers are already waiting	A message "You already have 5 offers waiting for the shop. Wait for an answer before sending more." is displayed			
	4	Consistency Check					
			Check if the portal lists only this supplier's orders: sign in as Supplier	The portal lists only this supplier's orders			
			Check if only this supplier's products are listed: look at Products you supply	Only the products the shop buys from this supplier are listed			
			Check if another supplier's order is listed: look for an order the shop placed with another supplier	It is not listed, and opening its id directly answers "not found"			
			Check if a confirmed order can be confirmed again: look at an order that is already confirmed	It has no Confirm order button			
			Check if a supplier with no orders is not rated: open the portal when no order was received in the last 12 months	A message "Not rated yet: no order from this supplier was received in the last 12 months." is displayed			
	5	Database Check					
			Check if the contact details are saved: click Edit contact details, change them, then click Save details	"Your details were updated." is displayed and the new values are saved			
			Check if the price is the current one: click Change price and enter the current price	products.cost_price already equals the price typed, so HTTP 400 "That is already the current price." is displayed and no price_proposals row is inserted			
	6	System Check					
			Check if the stars match the shop's: look at the	The same stars and conditions as the shop's			

MODULE	NO.	TESTCASE	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	COMMENT	STATUS
			star rating on the portal	Performance window are shown, with whether it meets the standard (4 stars or more)			
			Check if a price proposal reaches the shop: click Change price and enter a price different from the current one	A price_proposals row is inserted with current_price, proposed_price and status 'pending', the shop gets a notification "Price change proposed", and products.cost_price stays the same until the shop approves			